Mechanism-constrained deep learning recovers individual pharmacokinetics under undernutrition and water-toxicant co-exposure

An in silico study

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Declarations

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Data and code availability. All code, trained weights for all nine training runs, and the evaluation outputs reproducing every number in this manuscript are openly available at [REPOSITORY URL / DOI]. No empirical dataset exists or is required: the synthetic dataset regenerates deterministically from a fixed seed in under one minute, and the complete pipeline reproduces every figure and table in approximately 2.5 hours on two CPU cores. The diagnostic reported in “Sex: the model attributes more than the mechanism contains” is reproduced by `scripts/diagnose_sex.py`.

Author contributions. Pranav Manne conceived the study, wrote the simulator, model and analysis code, performed all experiments, and wrote the manuscript.

Supporting information. Supplementary figures S1–S2, the full pharmacodynamic target audit, per-drug and per-stratum dosing tables, and the threshold sensitivity analyses are provided in the accompanying Supporting Information.

Abstract

We show that a neural network constrained by compartmental pharmacokinetic structure can recover individual drug disposition from the sparse, noisy, partially missing measurements obtainable in a low-resource setting — in a simulated population whose generative process we specified.

Children with severe acute malnutrition are dosed in milligrams per kilogram, a rule that treats every kilogram of body mass as pharmacokinetically equivalent. Undernutrition violates that assumption along at least six axes simultaneously, and in opposite directions for different drugs: body composition shifts toward water, visceral organs are spared relative to body weight, inflammation suppresses cytochrome P450 3A4, hypoalbuminaemia raises the unbound fraction of acidic drugs while a rising acute-phase α1-acid glycoprotein lowers it for basic drugs, enteropathy blunts absorption, and cardiac output falls. These children

are also largely absent from the trials that set the labels, so the evidence needed to correct the dose does not exist. They are, in addition, disproportionately exposed to contaminated drinking water — an exposure that is not independent of their nutritional state but mechanistically coupled to it.

We constructed a generative simulator in which nutritional status and drinking-water toxicant exposure jointly determine body composition, organ mass, enzyme capacity, protein binding and gut integrity, and from these derive the pharmacokinetic parameters of nine essential medicines. We then trained a model on 36,000 simulated concentration–time records from 12,000 virtual patients. The model receives only observable covariates; it predicts seven physiological parameters, and a differentiable compartmental solver converts them to a concentration–time curve. Allometric exponents are fixed rather than learned, so any body-size effect the network acquires is by construction a departure from the allometric rule. Dose is excluded from the parameter encoder, making predictions exactly dose-proportional.

Against noise-free simulated truth, the model achieved a median fold error of 1.050 (98.6% of predictions within twofold; R^2 on the log scale 0.997) and recovered the latent pharmacokinetic parameters, never supplied as inputs, with R^2 between 0.981 and 0.999. In a matched comparison holding inputs, supervision and parameter count constant, replacing the mechanistic solver with a direct curve-regression head cost 1.54-fold in log-RMSE. Auxiliary supervision on parameters and latent physiology was worth considerably more than the solver: 3.72-fold. Ablation identified drinking-water chemistry as the dominant observable block: removing it degraded accuracy 2.05-fold, against 1.34-fold for removal of the entire bedside biochemistry panel.

Two negative results bound the method. The model did not transfer to a drug held out entirely (median fold error 11.2, 17.8% within twofold), and the failure localises to the held-out drug with no chemical analogue in the training set. Monte Carlo dropout was substantially overconfident, its nominal 90% interval covering 45.0% of the truth; uncorrected, this propagated into systematically poor dose decisions.

Comparing the simulator with published pharmacokinetic measurements in malnourished patients exposed two structural defects that the unit-test suite had not caught — an organ-sparing mechanism that was silently inert, and toxicant potencies set on an uninterpretable scale that implied guideline-compliant drinking water would cost a child a third of their glomerular filtration rate. After correction, predicted gentamicin clearance and distribution volume in severe malnutrition fall within published ranges, and a refitted absorption model reproduces published day-7 lumefantrine concentrations in both malnourished and well-nourished arms. One disagreement survives recalibration: the simulator predicts a halving of rifampicin exposure that two published studies do not support, and the discrepancy localises to plasma-protein binding.

These results establish that a mechanism-constrained architecture can invert a known, complex generative process from realistically degraded observations. They are not evidence about human pharmacokinetics. Calibrating the simulator against real sparse pharmacokinetic data from undernourished cohorts, and prospective validation, would both be required before any clinical interpretation is warranted.

Introduction

The populations in which drug dosing is least evidence-based are the ones in which physiology departs furthest from the reference patient. Severe acute malnutrition is the clearest case. Weight-band and milligram-per-kilogram dosing carry an implicit model: that drug clearance and distribution scale with body mass in a fixed, nutrition-independent way. That model is contradicted by the physiology of undernutrition on several fronts at once.

The contradictions do not point in a single direction, which is why a uniform correction factor cannot resolve them. Wasting removes adipose tissue faster than lean tissue, so total body water per kilogram rises: hydrophilic drugs gain distribution volume per kilogram while lipophilic drugs lose it. Liver and kidney mass fall sub-proportionally to body weight, so clearance per kilogram rises even as absolute clearance falls. Protein-energy deficit and the accompanying inflammatory state suppress some metabolic routes while sparing or inducing others, re-weighting the pathways by which a drug is eliminated. Hypoalbuminaemia raises the unbound fraction of acidic drugs; the acute-phase rise in α1-acid glycoprotein lowers it for basic drugs. Enteropathy reduces absorptive surface, raises gastric pH and depletes bile salts, which matters enormously for a lipophilic drug and barely at all for a small hydrophilic one. The net effect on exposure is therefore drug-specific in both magnitude and sign.

The obvious remedy, measure the pharmacokinetics in this population and re-derive the dose, is obstructed by the same conditions that create the problem. Dense sampling is rarely feasible in a stabilisation ward; laboratory panels are intermittent; serum creatinine, the usual route to renal function, is confounded precisely because muscle mass is low, so creatinine-based estimates are biased upward in the patients of interest. The available data are sparse, opportunistic and partially missing, which is the regime in which conventional population pharmacokinetic modelling is most dependent on structural assumptions and least able to test them.

A second exposure compounds the first, and the two are coupled rather than additive. Undernourished children are disproportionately likely to drink contaminated water, and nutritional deficit alters how that exposure is handled. Iron deficiency up-regulates divalent metal transport, which also carries lead, cadmium and manganese, so the same water yields a larger absorbed dose. Methyl-donor limitation under protein deficit slows arsenic methylation, so the same intake yields a larger and more toxic internal burden. The resulting body burdens then act back on drug disposition: cadmium is nephrotoxic and reduces renal clearance, arsenic injures hepatocytes while inducing selected enzymes, and lead impairs haem synthesis — haem being the prosthetic group of every cytochrome P450. Water toxicity is therefore not a confounder to be adjusted away but a second, partially independent axis of pharmacokinetic variation that interacts multiplicatively with the first.

This motivates a specific methodological question, which is the one this paper addresses. Suppose the generative process is as complex as the physiology above suggests. Can a model recover an individual’s pharmacokinetics from the degraded observations a real setting affords — and does embedding pharmacokinetic structure in the model help, or does a sufficiently large unconstrained network do as well?

We answer this as a controlled recovery experiment. Because no dataset exists in which the true individual pharmacokinetic parameters are known alongside full covariate information, we specify the generative process ourselves, degrade its output to resemble field conditions, and ask what a learned model can recover. This is the standard design for evaluating an inference method when ground truth is otherwise inaccessible, and it carries the standard caveat: it measures fidelity to the specified process, not to reality. We state that constraint here rather than in a closing paragraph because it governs the interpretation of every number that follows.

Relation to existing work

Mechanistic modelling of drug disposition in malnourished children is not new. A physiologically-based pharmacokinetic framework for this population has been published, combining malnourished-versus-healthy body composition with child-versus-adult physiology and evaluated against clinical reference data [5]. A systematic review of 49 studies covering 29 drugs documents what has actually been measured: malnutrition

altered pharmacokinetics in 21 of the 29, with reduced bioavailability for lumefantrine and several antiretrovirals, and increased elimination for albumin-bound drugs including rifampicin [6].

The contribution here is not the forward mechanistic model. It is threefold. First, a coupled generative model in which nutritional and environmental-exposure physiology jointly determine drug disposition, with every relationship stated as an inspectable function rather than fitted to data. Second, an architecture that runs the inverse direction — a neural encoder predicts physiological parameters from bedside-observable covariates and a differentiable compartmental solver produces the curve, with allometry fixed and dose excluded from the encoder, so that the learned component captures only departures from the conventional scaling rule. Third, an evaluation that reports where the method fails — transfer to an unseen drug, and uncertainty calibration, with the same prominence as where it succeeds.

A coupled generative model of nutrition, exposure and disposition

Identical milligram-per-kilogram dosing produces systematically different exposure across nutritional strata, and the direction of the effect differs by drug, which is the property that makes a single correction factor insufficient and a covariate model necessary.

The simulator proceeds causally rather than by regression. Anthropometry and oedema grade determine body composition; composition and a sparing exponent determine organ mass; organ mass and maturation determine glomerular filtration; protein deficit, inflammation and micronutrient status determine enzyme activity factors; plasma protein concentrations determine unbound fraction through the standard protein-ratio relation [24]; enteropathy determines mucosal surface, gastric pH, bile-salt availability and transit. Pharmacokinetic parameters are then derived from these quantities using a well-stirred hepatic model [7] and filtration-based renal clearance, not fitted. Two scaling constants per drug anchor the absolute magnitude to a declared reference clearance and volume in a well-nourished reference child, so that all patient-specific variation arises from the mechanism. Full specification is in Methods.

Drinking-water exposure enters the same causal chain at three points, which is what makes it a distinct axis rather than a correlate of poverty. Nutritional deficit modifies uptake, because iron deficiency up-regulates divalent metal transport shared by lead, cadmium and manganese [14]. It modifies detoxification, because arsenic methylation is S-adenosylmethionine dependent [21] and metallothionein synthesis is cysteine- and zinc-dependent, so the same intake yields a larger internal burden. The resulting burdens then modify disposition through saturating exposure–response relations: tubular toxicity reduces filtration, hepatocellular injury reduces functional enzyme mass, impaired haem synthesis reduces all cytochrome P450 activity proportionally [15], and competitive displacement raises unbound fraction further in an already hypoalbuminaemic child.

The drug library spans nine essential medicines chosen to dissociate these mechanisms rather than to sample a therapeutic class: a pure extracellular-fluid marker cleared almost entirely by filtration (gentamicin); a highly albumin-bound cephalosporin in which hypoalbuminaemia raises both free fraction and clearance (ceftriaxone); two drugs whose absorption depends on bile-salt solubilisation (rifampicin, lumefantrine); a poorly soluble high-first-pass compound in which absorption and first-pass metabolism move in opposite directions under malnutrition (albendazole); and others spanning renal and hepatic elimination.

Simulated under label dosing, the consequences separate by stratum and by drug. Exposure falls roughly fourfold for rifampicin and more steeply for lumefantrine in the most severe strata, driven by collapsed bile-salt-dependent absorption compounded by reduced metabolic capacity — while gentamicin exposure *rises* with oedema, because its distribution volume tracks an expanding extracellular compartment, and

ceftriaxone exposure falls as hypoalbuminaemia accelerates its clearance. Direction and magnitude are both drug-specific.

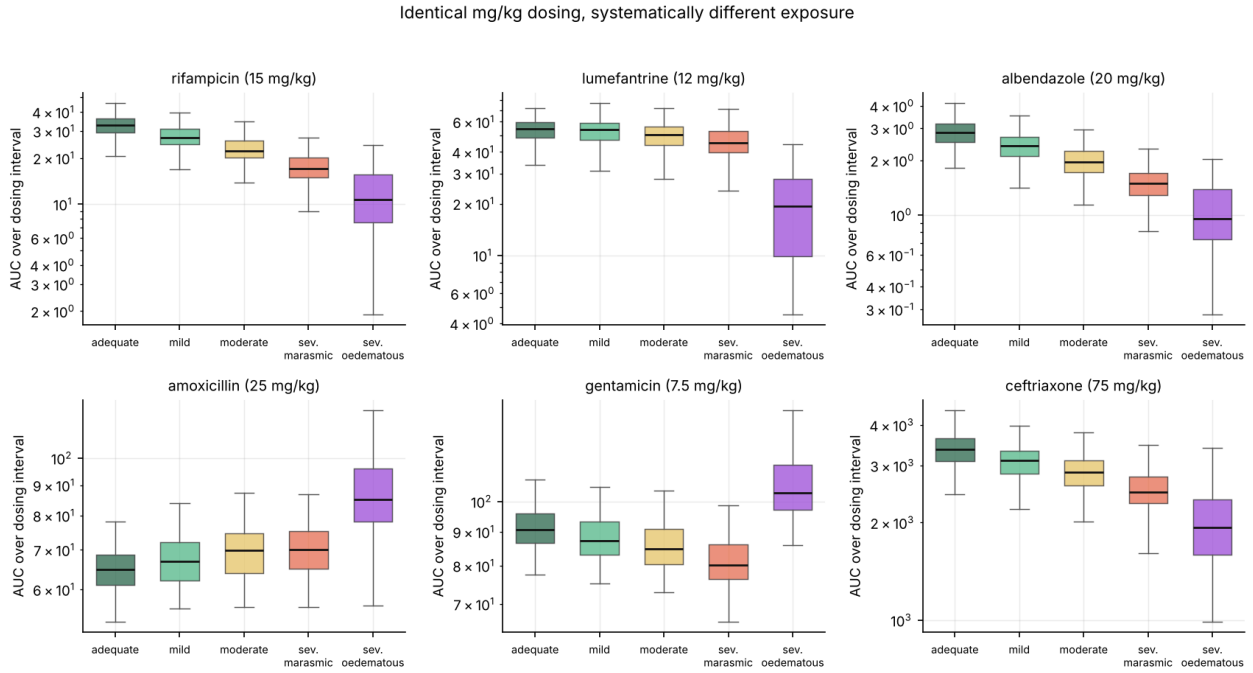


Figure 1 | Simulated exposure under identical label dosing diverges by nutritional stratum, and in opposite directions for different drugs. Area under the concentration–time curve over one dosing interval; 90 virtual patients per stratum per drug. All data synthetic.

The qualitative pattern for rifampicin and lumefantrine corresponds to a documented clinical concern in undernourished children. Here it is an emergent property of the specified mechanism rather than a fitted result, which makes it a face-validity check on the simulator's construction — not independent evidence for the phenomenon, since the mechanism that produces it is one we wrote.

External validation against published data

Comparing the simulator with published pharmacokinetic measurements found two structural defects that had survived the unit-test suite, showed one mechanism to be roughly threefold too strong, and left one disagreement that recalibration could not remove. This section reports that exercise, because it is the only part of this work that tests the simulator against something other than itself.

Four sources were used. Each was retrieved and read; no value was reconstructed from memory, and quantities that could not be obtained were left absent rather than estimated.

Source	Population	What it provides
Seaton et al. 2007 [1]	34 children with severe malnutrition, Kenya	Population PK of gentamicin: CL, V1, V2, Vss per kg
Denoeud-Ndam et al. 2016 [2]	255 children 6–59 months, Mali and Niger	Day-7 lumefantrine, SAM vs non-SAM
te Brake et al. 2015 [3]	36 adults with TB, Indonesia	Rifampicin AUC, CL, unbound fraction by nutritional status
Ngcobo 2025 [4]	Review	Directional findings across several drugs

Two structural defects

Organ sparing was inert. The simulator applied a sparing exponent to the weight-deficit ratio, but multiplied it onto body surface area computed from *actual* weight. Actual surface area already shrinks with the lost weight, so the two cancelled: kidney mass per kilogram moved from 13.3 to 13.4 g/kg across a twofold range of wasting. The mechanism the code appeared to implement was not operating at all, and no unit test caught it because each component behaved as specified in isolation. Building organ mass from the size the child would have at their own height without wasting, then applying the deficit exponent, restores the intended behaviour: 13.4 to 18.9 g/kg across the same range.

This correction is conceptually the same one the published PBPK framework [5] applies, which adjusts organ ratios for height against reference values rather than deriving them from overall weight loss. The convergence is reassuring and was arrived at independently.

Toxicant potencies were set on an arbitrary scale. Body burden is an internal quantity whose magnitude depends on assumed volume of distribution and half-life, so half-maximal-effect constants expressed in those units could not be sanity-checked — and were wrong by about two orders of magnitude. The simulator implied that a child drinking water meeting every WHO guideline would lose roughly 31% of their glomerular filtration rate. Re-expressing every potency as a multiple of that toxicant's guideline-equivalent steady-state burden makes the claims legible and corrects the scale: median predicted tubular toxicity is now 0.3% for clean municipal water, 3% peri-urban, and 11% for industrial runoff.

Together these defects had pushed predicted gentamicin clearance to 0.025 L/h/kg against a published 0.103 — fourfold too low, and in the wrong direction, since the published value is within the normal paediatric range.

Gentamicin: the cleanest mechanistic test

Gentamicin is an extracellular-fluid marker cleared almost entirely by filtration, so it probes the two mechanisms the simulator most directly claims. After the structural corrections and after anchoring the drug's reference clearance and volume to published normal paediatric values, the comparison is:

Parameter	Simulator	Seaton et al.	Fold difference	Within published range
CL (L/h/kg)	0.088	0.103 (0.02–0.16)	1.17	yes
Vss (L/kg)	0.509	0.670 (0.26–1.31)	1.32	yes
V2 (L/kg)	0.346	0.406	1.17	—
V1 (L/kg)	0.163	0.254	1.56	—

The informative quantity is not the absolute agreement, which partly reflects the chosen anchors, but the ratio: the simulator now predicts weight-normalised renal function in severe malnutrition at 1.00 times its well-nourished reference, which is what the published value implies. The central compartment remains 1.56-fold too small and is the clearest remaining structural discrepancy.

Lumefantrine: a mechanism refitted

The simulator predicted a day-7 lumefantrine ratio of 0.28 between children with and without severe acute malnutrition, against a published median ratio of 0.75. Its oral-absorption penalty was roughly threefold too severe.

The enteropathy coefficients acting on mucosal surface and bile-salt availability were therefore refitted to this target, leaving the functional forms unchanged. Predicted absolute concentrations fall inside the

published interquartile range in both arms (SAM 173 against a published median 246, IQR 160–438; non-SAM 290 against 330, IQR 216–503).

Two caveats constrain what this fit means. The published contrast is itself confounded: the malnourished arm was younger (17 against 28 months) and vomited more often, both of which lower exposure independently of nutritional status, so 0.75 is an upper bound on the malnutrition effect and the fitted coefficients inherit that bias. And the fit is noisy. A first attempt matched a single simulated cohort to 0.744 against the published 0.745, which looked like an excellent result and was not: the same coefficients produced ratios from 0.51 to 0.69 on other random draws. Averaging over independent draws gives a between-draw standard deviation of 0.063, so agreement closer than about 0.06 on this target carries no information.

Rifampicin: a disagreement that stands, and is narrower than it looks

The simulator predicts that severe malnutrition reduces rifampicin exposure to 0.48 of its well-nourished value. Two published studies report no significant difference: te Brake et al. found an AUC ratio of 1.00 in adults [3], and a review attributes to Mukherjee et al. a finding of no significant difference in Cmax or AUC in children with severe malnutrition [4].

Decomposing the predicted ratio locates the disagreement: bioavailability falls to 0.63 of reference while weight-normalised clearance *rises* to 1.34, and the clearance rise is driven mainly by hypoalbuminaemia raising the unbound fraction by 29%. te Brake et al. measured unbound fraction directly and found it identical at 10.5% in both groups despite lower albumin, which contradicts that pathway. Their albumin difference was narrow (3.3 against 3.8 g/dL) and not itself significant, so the study is underpowered to exclude a binding effect of the size the simulator predicts over the much wider albumin range of paediatric severe malnutrition — but it is the only direct measurement available, and it does not support the mechanism.

The systematic review [6] complicates this in the simulator's favour. Across 49 studies it reports *increased* elimination in malnutrition for albumin-bound drugs including rifampicin, which is the direction the simulator predicts for the clearance limb. What is contested, then, is the magnitude of the binding effect and whether unbound fraction is the responsible pathway — not obviously the sign of the clearance change. The bioavailability limb, which contributes more to the predicted AUC ratio, remains untested against this evidence.

We have not refitted any of this. Forcing a match to adult data, for a phenotype that differs from paediatric kwashiorkor, would substitute one unsupported claim for another. The disagreement is recorded, the responsible pathway is identified, and the relevant test in the suite is annotated as a characterisation of current behaviour rather than an assertion that it is correct.

One directional check fails

Of two directional predictions testable against review-reported findings, one holds and one does not. Aminoglycoside distribution volume rises in malnutrition in the simulator (ratio 1.19), consistent with reports of expanded streptomycin volume in kwashiorkor. Beta-lactam weight-normalised clearance should fall, following reports of reduced renal clearance of cefoxitin and penicillin in protein-energy malnutrition; the simulator predicts a slight rise (ratio 1.04), because the organ-sparing correction holds filtration per kilogram flat while hypoalbuminaemia raises the unbound fraction. The same binding pathway implicated in the rifampicin disagreement is implicated here, which makes plasma-protein binding the most probable location of the simulator's largest remaining error.

What this changes

The simulator is better constrained than it was, and in two respects it is now anchored to measurement rather than to plausibility. It is not validated. Three drugs have been compared on a handful of quantities, one mechanism has been fitted to a single confounded study, and the largest drug class in the library has not been tested at all.

The effect on the rest of this paper is limited but real. The machine-learning results are statements about recovering a generative process, and that argument is unaffected by whether the process matches reality. What changes is the standing of the simulator's own outputs: the exposure divergence by stratum now rests partly on measurement for gentamicin and lumefantrine, and rests on a contested mechanism for rifampicin, where the magnitude shown should be read as an upper bound.

A mechanism-constrained architecture

The network never predicts a concentration. It predicts seven physiological parameters, and a differentiable compartmental solver converts them into a concentration–time curve by exactly solving the same linear system the simulator used.

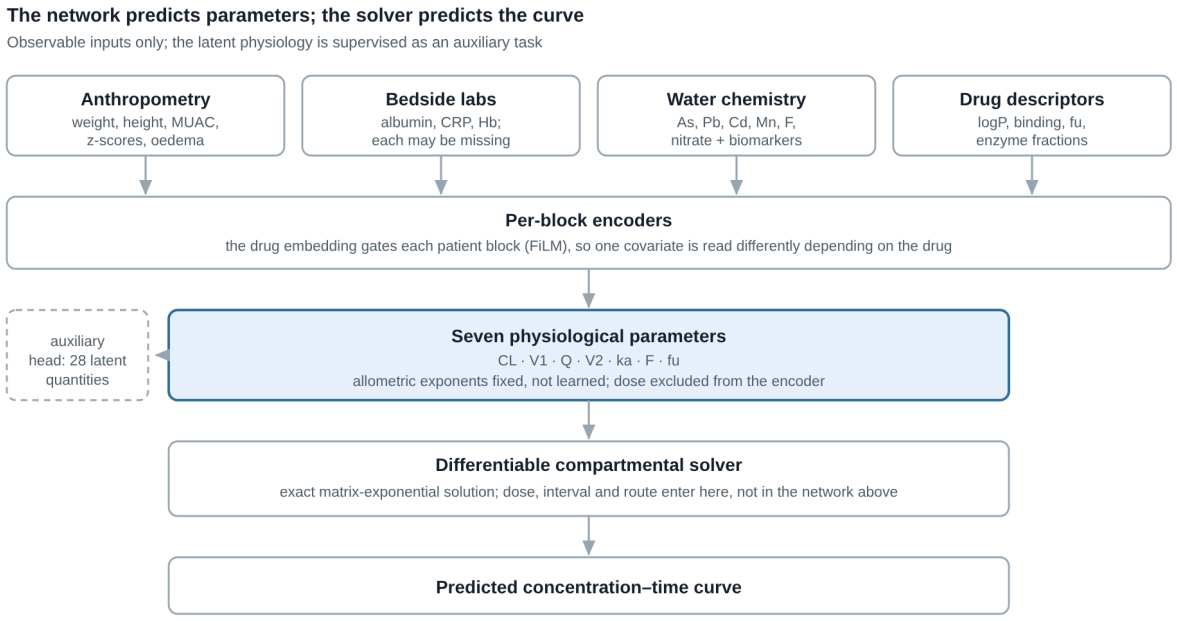


Figure 2 | The learned component predicts parameters; the fixed component predicts the curve. Inputs are restricted to quantities obtainable at or near the bedside. Latent physiology — organ mass, filtration rate, enzyme activity, unbound fraction — is withheld from the input and instead predicted as an auxiliary task.

Inputs are restricted to quantities obtainable at or near the bedside: anthropometry; a small bedside biochemistry panel, each value independently allowed to be missing; drinking-water chemistry from a field test kit; two internal exposure biomarkers; and the drug's own physicochemical descriptors. Latent physiology — organ mass, filtration rate, enzyme activity, unbound fraction — is withheld from the input and instead predicted as an auxiliary task.

The separation of observable from latent is a deliberate constraint on the learning problem. The simulator knows each virtual child's liver mass, glomerular filtration rate and cytochrome P450 activity; a clinician does not. Supplying those quantities as inputs would make recovery trivial and the result uninformative.

They are therefore withheld and supervised as a side task, which both regularises the shared representation toward a physiologically meaningful space and renders the model interpretable: one can ask why it expects under-exposure and receive an answer in terms of inferred enzyme activity and bile-salt-dependent absorption rather than feature attributions. As the results below show, this choice turns out to carry more of the model's performance than any other.

Three structural constraints distinguish the architecture from a conditional curve regressor.

Parameters are decomposed into a drug-level typical value and a patient-level departure. The log-parameter vector is the sum of a term computed from physicochemical descriptors alone, a fixed allometric term in log body weight, and a learned patient-specific deviation. The descriptor branch can place a drug it has not seen; the patient branch then corrects. This mirrors how a pharmacologist reasons, and it is the only route by which transfer to a new compound could occur at all.

Allometric exponents are fixed rather than learned, at 0.75 for clearance terms and 1.0 for volumes. These are not free parameters. Consequently every body-size effect the network acquires is, by construction, a departure from the allometric rule, which is precisely the quantity of interest, because the allometric rule is what milligram-per-kilogram dosing already assumes.

Dose is excluded from the parameter encoder. Clearance and volume are properties of the patient and the drug, not of the dose administered. Excluding the regimen makes the predicted profile exactly proportional to dose — verified to ten significant figures, including at doses outside the training range, so the model cannot silently learn a dose-dependent clearance and then be asked to extrapolate. This is what makes the inverse problem well-posed rather than merely tractable.

A fourth choice concerns how covariates are combined. A patient covariate has no drug-independent meaning: serum albumin dominates ceftriaxone disposition and is irrelevant to gentamicin; bile-salt deficiency destroys lumefantrine absorption and barely touches amoxicillin. The drug embedding therefore modulates the patient encoder's features multiplicatively (feature-wise linear modulation [20]) rather than being concatenated to them, so the same measurement is read differently depending on the compound in hand.

Missingness is handled explicitly rather than by imputation. Each laboratory value carries an observed-indicator channel, and absent values are replaced by a learned per-feature token, so that "albumin not measured" and "albumin measured as low" remain distinct inputs. The model is trained with missingness present at field-plausible rates. A model that silently requires a serum albumin is a model that cannot be deployed where it would be needed.

Recovery of latent pharmacokinetics, and the contribution of structure

The model recovered the noise-free simulated curve to a median fold error of 1.050, with 98.6% of predictions within twofold and an R^2 on the log scale of 0.997, from sparse noisy samples with partially missing laboratory data.

Accuracy is reported against the noise-free generative curve at all 50 time points, while the model observed only the noisy, sparsely sampled subset — two to four opportunistic samples in 55% of records. Error is expressed as fold error, the ratio of predicted to true concentration, because an absolute error means something different at a trough and at a peak. Twofold is the conventional acceptability threshold in pharmacometric prediction.

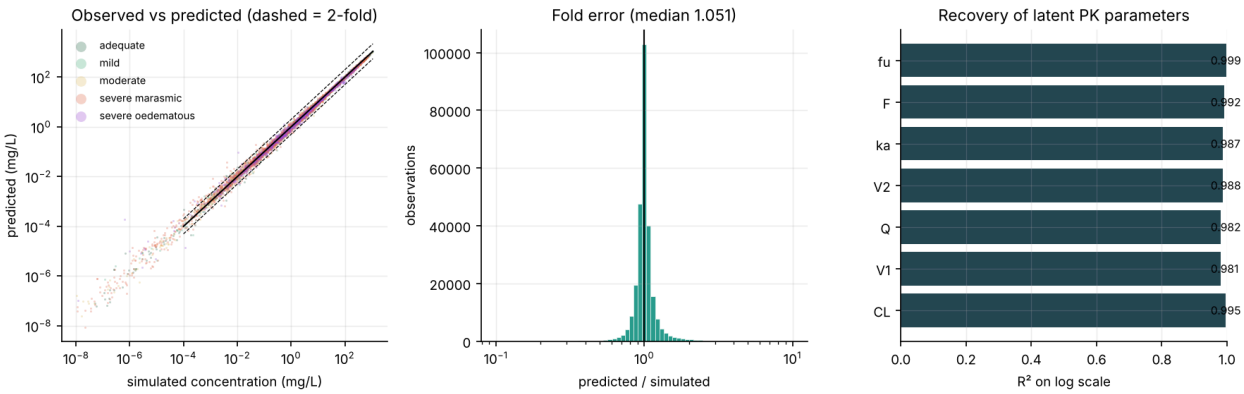


Figure 3 | Recovery against noise-free simulated truth. Left, predicted against simulated concentration coloured by nutritional stratum; dashed lines mark twofold. Centre, distribution of fold error. Right, recovery of the seven latent pharmacokinetic parameters, which are never supplied as inputs. Held-out patients, 5,400 records. All data synthetic.

Isolating the contributions of solver and supervision

Comparing a hybrid model against a black box is uninformative unless inputs, supervision and capacity are matched. The hybrid receives auxiliary supervision on parameters and latent physiology that a curve regressor cannot use, so a naive comparison confounds architecture with supervision. We therefore trained all four cells of the design: both architectures, each with and without the auxiliary targets.

Model	Supervision	Parameters	log-RMSE	Median fold error	Within twofold
Hybrid	full	678,707	**0.190**	1.050	98.6%
Hybrid	concentrations only	658,583	**0.705**	1.155	93.0%
Black box	full	660,628	0.979	1.177	91.1%
Black box	concentrations only	640,504	**1.087**	1.214	90.0%

The controlled architectural contrast is the second and fourth rows: identical inputs, identical supervision (noisy sparse concentrations and nothing else), parameter counts within 3%. **The mechanistic solver is worth a factor of 1.54 in log-RMSE.**

The controlled supervision contrast is the first and second rows: identical architecture, differing only in whether the latent targets are supplied. **Auxiliary supervision is worth a factor of 3.72**, more than twice the solver's contribution.

This reverses the emphasis of an earlier version of this work, which reported the solver at 2.47-fold against roughly 6% from supervision and concluded that encoding known physics as architecture beat supplying it as training signal. Against the corrected simulator that conclusion does not hold, and the change is worth stating plainly rather than quietly updating.

The reason is diagnosable. Softening the absorption model during external calibration made bioavailability and clearance harder to separate from concentration data alone. This penalises the hybrid specifically: without parameter supervision its seven-parameter bottleneck becomes partly unidentifiable, and its parameter-recovery error rises sharply even as its curve predictions stay good. The black box has no bottleneck to be confused about. Part of the hybrid's original margin therefore came from a simulator in which the parameters were easier to identify than they should have been.

What survives is a narrower and better-supported claim. The solver still contributes a real and measurable 1.54-fold, and it buys four properties that no amount of supervision provides: every prediction is a solution of a mass-balance system with non-negative parameters; predictions are exactly proportional to dose, including outside the training range; the bottleneck is seven quantities a pharmacologist recognises; and two or three opportunistic samples suffice to constrain seven parameters where they would come nowhere near constraining a free-form curve. The black box reaches 90.0% within twofold and is not catastrophically bad; it simply cannot be asked to do the inverse problem reported below.

The broader design question, whether to encode known structure as architecture or supply it as training signal, does not have a single answer here. Architecture constrains the model everywhere, including at dose levels and covariate combinations the training set never contained; supervision constrains it only where data exist. In this setting supervision was worth more numerically, and architecture was worth more structurally. Both were needed: the full model is 5.7-fold better than the black box trained without either.

Recovery of the latent parameters

Parameter	R^2 (log scale)	Median fold error
Unbound fraction (fu)	0.999	1.021
Clearance (CL)	0.995	1.051
Bioavailability (F)	0.992	1.071
Peripheral volume (V2)	0.988	1.061
Absorption rate (ka)	0.987	1.057
Intercompartmental clearance (Q)	0.982	1.052
Central volume (V1)	0.981	1.052

Unbound fraction is recovered most precisely, which is expected: it is determined almost entirely by measured albumin through a deterministic relation, so when albumin is observed the mapping is close to a lookup. Central volume is recovered least precisely, also expected: with sparse sampling the early distribution phase is rarely captured, and V1 is identifiable mainly from early time points.

Accuracy across nutritional strata

An aggregate metric would conceal the failure mode that matters most here — a model accurate on average and poor in severe disease would have failed at its stated purpose.

Stratum	Records	Median fold error	Within twofold
Adequate	1,212	1.046	99.4%
Mild	225	1.054	98.4%
Moderate	912	1.050	99.1%
Severe, marasmic	2,511	1.049	98.2%
Severe, oedematous	540	1.063	98.5%

Accuracy is essentially flat across severity, degrading slightly in oedematous malnutrition — the stratum in which the physiology departs furthest from the well-nourished reference, and in which the simulator itself places the most variance.

Which observations carry the signal

Drinking-water chemistry was the most load-bearing input block: removing it degraded accuracy 2.05-fold, against 1.34-fold for removal of the entire bedside biochemistry panel.

Ablations zero an input block while leaving architecture and capacity unchanged, so any difference is attributable to information rather than to model size. Each ablated model was retrained from initialisation.

Information removed	log-RMSE	Degradation	Within twofold
None (full model)	0.190	—	98.6%
Bedside biochemistry panel	0.254	1.34×	97.5%
Drinking-water chemistry	0.389	2.05×	93.7%
Both (anthropometry only)	0.415	2.19×	93.4%

The asymmetry is interpretable, and it is not a claim that toxicant exposure matters more than nutritional status. Anthropometry partially proxies for the biochemistry panel: a wasted, oedematous child is likely to be hypoalbuminaemic and inflamed, so the clinical examination recovers much of what the laboratory would have told us. Nothing in the clinical examination proxies for what is in the well. Removing water chemistry therefore leaves a large, structured source of variance entirely unobservable, whereas removing the laboratory panel leaves a partially redundant one.

That the two ablations are nearly equal in their combined effect (2.05× and 2.19×) confirms the redundancy directly: once water chemistry is gone, discarding the laboratory panel as well costs almost nothing further.

The magnitude of this asymmetry is smaller than earlier versions of this work reported (3.5× against 1.2×, from the pre-calibration simulator). The ordering and the redundancy structure are unchanged; the contrast is roughly half as sharp.

Covariate attribution

Permutation importance partly recovers the simulator’s own causal ordering without having been shown it.

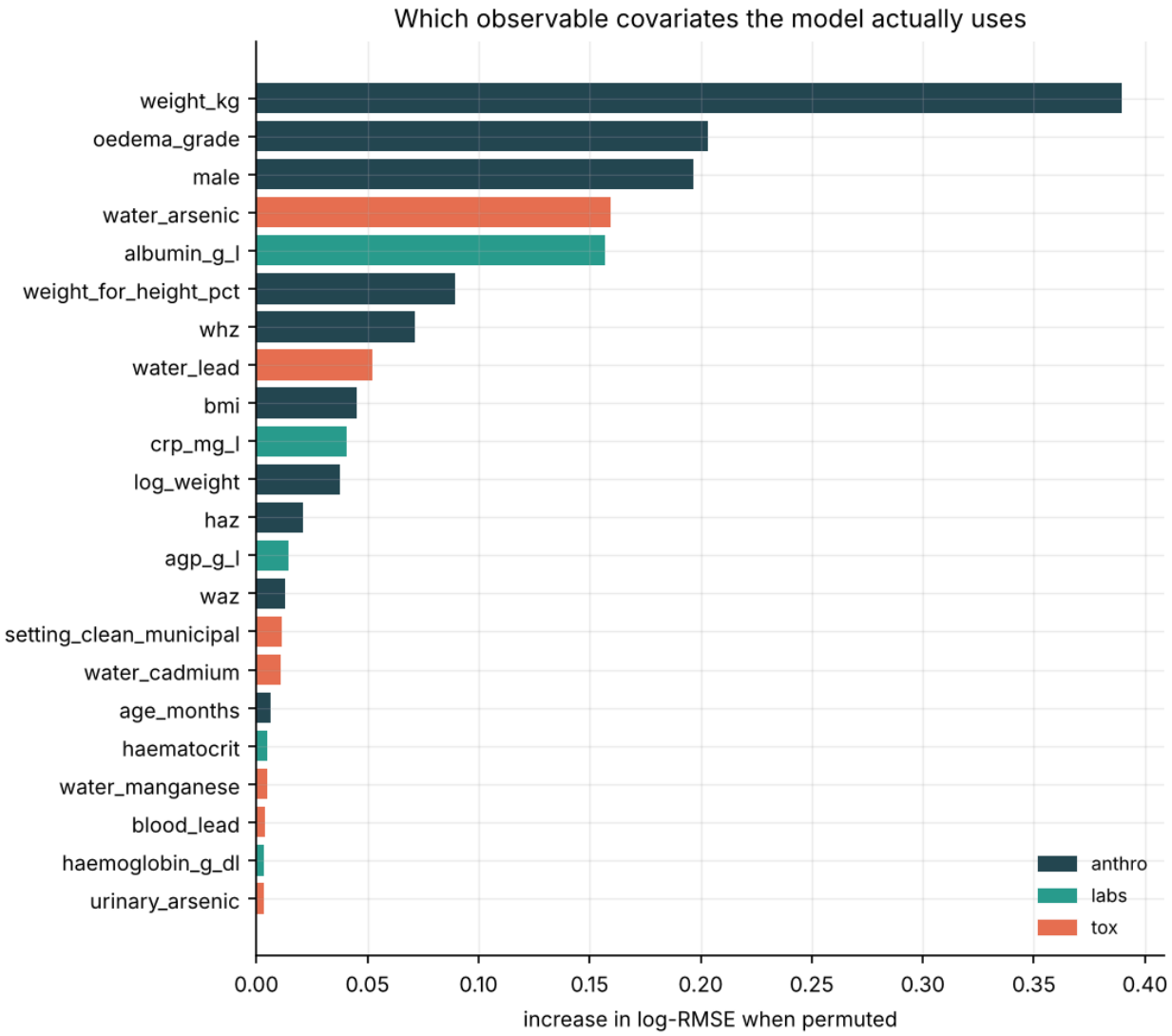


Figure 4 | Permutation importance on log-concentration RMSE. Each feature is permuted within its observed marginal, holding all else fixed; two repeats, 1,500 held-out records. Colour denotes input block. All data synthetic.

Rank	Feature	Block	Δ log-RMSE
1	body weight	anthropometry	0.389
2	oedema grade	anthropometry	0.203
3	sex	anthropometry	0.197
4	water arsenic	toxicants	0.160
5	serum albumin	labs	0.157
6	weight-for-height	anthropometry	0.090
7	weight-for-height z-score	anthropometry	0.071
8	water lead	toxicants	0.052

Permutation is the appropriate tool here rather than gradient attribution, because the inputs are strongly correlated and a gradient describes an infinitesimal perturbation the data never actually makes. Permuting within the observed marginal keeps each feature in a plausible range.

Body size sets the allometric scale; oedema reorganises the water compartments; arsenic acts on hepatic elimination through one of the strongest exposure–response relations in the simulator; albumin sets unbound fraction; lead acts diffusely through haem synthesis and is correspondingly weaker. Those placements reproduce the construction of the generative model.

Two features do not. Water cadmium, which ranked fourth in the pre-calibration model, now falls outside the top eight; the potency rescaling described above is the likely cause, since it reduced cadmium's tubular effect substantially. And sex ranks third, above every toxicant and every laboratory value, which the simulator's construction does not justify. We investigated this rather than leaving it as an anomaly, and the result is reported next.

Sex: the model attributes more than the mechanism contains

Sex enters the simulator at one place only, the sex-specific constants of the fat-free mass relation, which propagate to total body water and hence to distribution volume. Its causal magnitude is therefore measurable directly: flip sex on a sampled patient, rebuild the body composition that follows, re-derive the pharmacokinetic parameters, and hold anthropometry fixed so that only this pathway moves. Doing so changes fat-free mass by 3.3%. The model's own response to flipping the same input is obtained the same way, on held-out records.

Parameter	Simulator	Model	
CL	0.0000	0.0220	model responds; simulator does not
V1	0.0422	0.1476	3.5×
Q	0.0000	0.0196	model responds; simulator does not
V2	0.0417	0.1983	4.8×
ka	0.0000	0.0341	model responds; simulator does not
F	0.0000	0.0234	model responds; simulator does not
fu	0.0000	0.0307	model responds; simulator does not

RMS log change in the parameter when sex is flipped. The model moves the volumes three-and-a-half to nearly five times as much as the simulator does, and moves clearance, intercompartmental clearance, absorption rate, bioavailability and unbound fraction, none of which the simulator moves at all.

The permutation importance of sex is therefore a true statement about the model and a false one about the mechanism: the model has learned a stronger and broader sex dependence than the generative process contains. The most likely reason is redundancy. Four other inputs, the three anthropometric z-scores and weight-for-height percentage, are computed against sex-specific references, so sex is partly recoverable from them, and with redundant inputs a network can distribute a learned effect among them in any proportion. This architecture does not prevent that: the seven-parameter bottleneck constrains what the intermediate quantities mean, not how the encoder arrives at them.

Two consequences follow. First, the attribution result should be read as a partial rather than a complete consistency check: it recovers the simulator's ordering for the mechanistic covariates and exposes an over-attribution for a covariate the simulator barely uses. Second, auxiliary supervision on latent physiology did not prevent the over-attribution, which bounds the interpretability claim made for it elsewhere in this paper. Reproduce with `python scripts/diagnose_sex.py`.

An earlier version of this work described this attribution as recovering "exactly the causal ordering the simulator encodes." That was an overstatement. It recovers most of it, with one prominent anomaly. This is a

consistency check rather than a discovery in either case: the model was trained on data generated by that ordering, so recovering it demonstrates that the attribution method and the architecture agree with known ground truth, which is what would make the same method worth trusting in a setting where ground truth is not known. The anomaly is a caution about how far that trust extends.

Where generalisation fails

The model extrapolates acceptably to an unseen exposure setting and fails outright on an unseen drug, and the asymmetry has a specific, diagnosable cause rather than being a general limit of the approach.

Random patient splits test interpolation within a population the model has seen. That is the easy question. We therefore constructed two harder splits, each withholding an entire region of the covariate space.

Split	Held out	log-RMSE	Median fold error	Within twofold
Random patients	—	0.190	1.050	98.6%
Exposure setting	all industrial-runoff water	0.855	1.437	74.9%
Drug	rifampicin and gentamicin	3.219	11.184	17.8%

Unseen exposure setting: degraded but usable

Trained without any patient drinking industrial-runoff water, the setting with the highest lead and cadmium, and tested exclusively on that setting, median fold error rose from 1.050 to 1.437 and twofold coverage fell to 74.9%. The model extrapolates along the toxicant axis reasonably well, because the exposure–response relationships it learned at lower burdens remain the governing relationships at higher ones; it is extrapolating along a curve it has partially traced, not into an unmodelled regime.

Unseen drug: the method does not transfer

Holding out two drugs entirely produced a median fold error of 11.2: a failure, not a degradation.

Held-out drug	Median fold error	Within twofold
Gentamicin	19.42	8.7%
Rifampicin	4.77	27.2%

The explanation is structural. Remove gentamicin from the library and no remaining training drug is simultaneously intravenous, essentially unbound and cleared almost entirely by glomerular filtration. Ceftriaxone is intravenous but 94% protein-bound; fluconazole is largely renal but orally administered and widely distributed. The drug encoder is asked to place a compound in a region of descriptor space containing no training examples, and it cannot. Rifampicin, which has close analogues in the training set — lipophilic, bile-dependent, cytochrome P450 3A4-cleared, overlapping substantially with lumefantrine and albendazole — degrades to a 4.8-fold error instead: poor, but recognisably in the right regime.

The conclusion is that transfer to a new compound occurs by interpolation within covered chemical space and not otherwise, and that a nine-drug library covers very little of that space. This bounds the method as it stands: it is a model of the drugs it was trained on, not a general pharmacokinetic predictor. Whether the descriptor branch would support genuine transfer given a library an order of magnitude larger is an open question this experiment cannot answer, and it is the most direct test of the architecture’s central claim that we have not performed.

We report this prominently because the held-out-drug result is the experiment most likely to have been omitted from a paper making the positive claim, and because the mechanism of failure, extrapolation outside covered descriptor space, is the one a user of such a model would most need to know how to detect.

Calibrated uncertainty, and why miscalibration changes decisions

The residual-error model was well calibrated without correction; Monte Carlo dropout was substantially overconfident, its nominal 90% interval covering 45.0% of the truth. Left uncorrected, this propagated into systematically poor downstream decisions, which is the reason the finding is reported as a result rather than a diagnostic.

The two uncertainty components answer different questions and must be scored against different targets. Residual uncertainty, emitted by a heteroscedastic head, describes what a *measurement* will do, so its target is the noisy observation. Epistemic uncertainty, estimated from the spread across Monte Carlo dropout samples [17], describes what the *underlying curve* does, so its target is the noise-free truth. Scoring the combined interval against the noise-free truth, the obvious procedure and the wrong one, double-counts assay error and makes any model appear conservative.

Nominal level	Total interval vs observations	Epistemic vs truth, raw	Epistemic vs truth, recalibrated
50%	49.4%	20.3%	54.0%
80%	79.4%	36.6%	78.3%
90%	89.3%	45.0%	86.0%
95%	94.2%	51.5%	90.5%

The residual model is close to nominal at every level. The dropout estimate is not, and the deficit is severe: a stated 95% interval contains the truth about half the time. This is a known property, dropout masks explore a far smaller function space than genuine posterior uncertainty over weights — but its magnitude here is large enough to invalidate any decision procedure that consumes the posterior.

We correct it with a single scalar fitted on the validation split and never on test: the factor that makes the standardised residuals have unit robust spread. The fitted value was 3.13, and applying it brings 90% coverage from 45.0% to 86.0%. This is variance recalibration, the uncertainty analogue of temperature scaling [18, 19], and it costs one parameter. A deep ensemble would be the better remedy; this is the cheap one, and its merit is principally that it is not silently overconfident.

The correction is not cosmetic

The dose-selection procedure described next consumes this posterior to compute a probability of attaining a pharmacodynamic target. With the uncorrected posterior, a dose the model rated at 90% confidence was in fact near the median of the predictive distribution, so the selected dose sat on the target with no margin and the realised attainment was far below the stated figure. In an early configuration this produced *lower* target attainment than the label dose — the model confidently selecting doses that failed.

This is a general point about mechanism-constrained models rather than an implementation detail. The architecture guarantees that predictions are physically admissible; it guarantees nothing about whether the predicted *distribution* is honest. Structural correctness and calibration are independent properties, and a model that has the first without the second is more dangerous than one that has neither, because its outputs are individually plausible.

Recovery of dose decisions

The final test of an inversion is whether it supports a decision. We therefore posed the inverse problem — given this simulated patient, what regimen? — and had the generating simulator, which the model saw only through noisy sparse samples, score the answer.

The design is deliberately adversarial toward the model. A fresh population of 120 virtual patients was drawn, disjoint from all training and validation data. For each patient and each of eight drugs, the model received only observable covariates degraded in the usual way, proposed a regimen, and the simulator then evaluated that regimen against its own pharmacokinetic–pharmacodynamic criterion. Label dosing was scored identically. Because parameters are dose-independent by construction, the dose–exposure map is exactly linear for area under the curve and monotone for time above a threshold, so the inverse problem is solved by bisection on a function guaranteed to be monotone rather than by gradient search.

This measures whether the inversion succeeded, not whether anything clinical was achieved. Both arms are scored by the same simulator that generated the training data, and against a target we also specified. A favourable result establishes that the learned surrogate inverted the generative process accurately enough to act on it. It is an upper bound on, and not evidence of, benefit in any real population.

	Label dosing	**Model-selected**
Reached the target	69.1%	**87.5%**
Exceeded the toxicity threshold	1.6%	**0.0%**
Median dose ratio	1.00×	1.00× (10th–90th pct 1.00–2.16×)

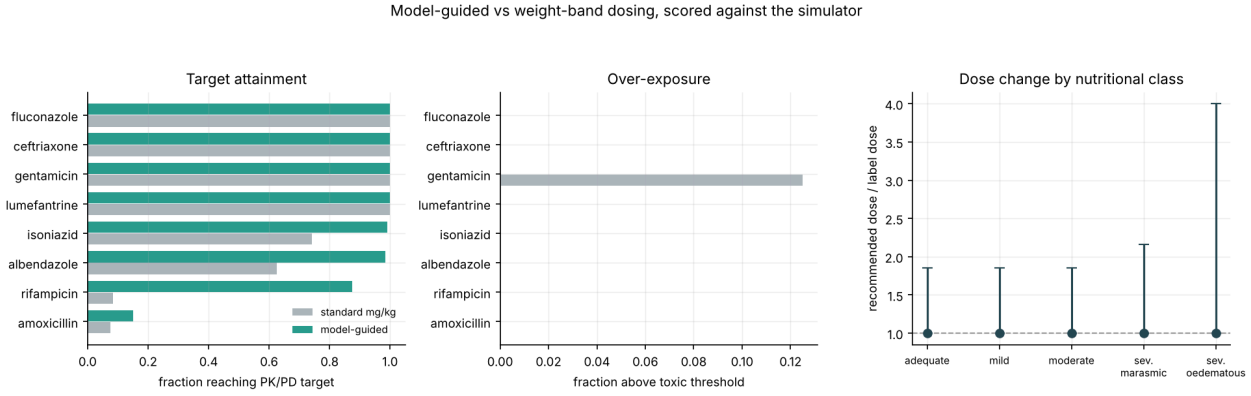


Figure 5 | Simulator-scored outcomes under label dosing and model-selected dosing. Left, fraction reaching the target. Centre, fraction exceeding the toxicity threshold — zero for every drug under label dosing after recalibration. Right, ratio of selected to label dose by nutritional stratum, with 10th–90th percentile range. 960 decisions across 120 virtual patients and 8 drugs. All data synthetic.

The median dose ratio of 1.00 is the informative number. The procedure does not shift the population dose; it leaves the label dose unchanged for patients it is already serving and moves only those it is failing. Aggregate attainment rises without a population-wide dose increase.

Drug	**Reached target**	**Exceeded toxicity**	**Median dose**	**Interval changed**
Rifampicin	8% → **88%**	0% → 0%	1.85×	4%
Albendazole	62% → **98%**	0% → 0%	1.17×	6%
Isoniazid	74% → **99%**	0% → 0%	1.17×	0%
Amoxicillin	8% → **15%**	0% → 0%	2.16×	0%

Drug	Reached target	Exceeded toxicity	Median dose	Interval changed
Gentamicin	100% → 100%	**12.5% → 0%**	1.00×	6%
Lumefantrine	100% → 100%	0% → 0%	1.00×	0%
Ceftriaxone	100% → 100%	0% → 0%	1.00×	0%
Fluconazole	100% → 100%	0% → 0%	1.00×	0%

The per-drug pattern tracks the mechanism. The three drugs with the largest attainment gains, rifampicin, albendazole and isoniazid, are those the simulator under-exposes most severely under label dosing. The drugs already attaining their target are left alone.

Stratum	Reached target	Dose ratio, 10th–90th pct
Adequate	74% → 87%	1.00–1.85×
Mild	72% → 88%	1.00–1.85×
Moderate	71% → 87%	1.00–1.85×
Severe, marasmic	65% → 87%	1.00–2.16×
Severe, oedematous	69% → **91%**	1.00–4.00×

The gain is largest in the most severe stratum, and the dose range widens monotonically with severity. In oedematous malnutrition the 10th-to-90th-percentile selected dose spans 1.00 to 4.00 times the label dose. That spread is the argument against a single weight band expressed as one number: within a stratum defined by a weight band, the regimens the simulator considers appropriate differ by fourfold.

The over-exposure result is governed by one unsourced threshold

Gentamicin carries the only toxicity signal in this experiment, and its size is set entirely by the trough threshold, which was not sourced in earlier versions of this work. The published once-daily paediatric target is ≤0.5 mg/L, and ≤1.0 mg/L for neonates [26]. The implementation previously used 2.0 mg/L, a conventional multiple-daily-dosing value with no paediatric source.

Gentamicin trough threshold	Over-exposed, label dosing	Over-exposed, model
2.0 mg/L (unsourced)	0.0%	0.0%
1.0 mg/L (neonatal [26])	2.5%	0.0%
0.5 mg/L (paediatric [26])	**12.5%**	**0.0%**

The population here is 6–59 months, so 0.5 mg/L applies, and that is the value now implemented; all three runs are retained in `outputs/ci_gent_*.json`. Efficacy is unaffected in every case, gentamicin attains its peak target in 100% of patients under both dosing arms, so the entire result is the toxicity column.

This is worth stating carefully, because an intermediate version of this work reported the opposite. After external validation corrected gentamicin's reference clearance and volume, over-exposure fell to zero and we recorded the earlier finding as an artefact of an uncalibrated simulator. That conclusion was premature: the finding had been suppressed by a toxicity threshold four times too lenient, and it returns once the threshold is sourced. The lesson is the one this section is about — a result that moves this much under a parameter nobody sourced is a statement about the parameter until the parameter is settled, in either direction.

The structural point stands and now has its demonstration back: the dosing interval is used in 6% of gentamicin cases, because a recommender restricted to dose cannot separate a peak-driven efficacy criterion

from a trough-driven toxicity criterion. Reducing the dose to rescue a toxic trough discards the peak that was doing the work.

Amoxicillin is determined by its target, not by the model

Amoxicillin attains 8% under label dosing and only 15% under the model, making it the one drug the procedure barely helps. But the implemented criterion is 50% fT>MIC, the near-maximal bactericidal target for penicillins [25], while published targets for the class span roughly 40–50%. Re-running the identical model and procedure with a 40% target gives **19% → 72%**.

The conclusion for this drug is therefore set by the target definition rather than by anything the model did. Two further checks make the general point sharper. Simulating a *well-nourished* reference child at label dose, the implemented target is not attained for amoxicillin (fT>MIC 0.29 against a target of 0.50) or for rifampicin (free AUC/MIC 24 against a target of 38). For those two drugs the low baseline attainment reported above is therefore not attributable to undernutrition at all: the target is not reached in the reference patient either. Conversely, the isoniazid target of 14 is met by a factor of roughly sixteen in a well-nourished child (free AUC/MIC ~232), against published AUC0-24/MIC targets of 567 and ~707, so its 74% → 99% improvement measures very little.

Three of eight drugs therefore have targets that determine their own results. This is the clearest instance of a general problem stated below.

The objective is a separate choice from the model

We also evaluated a dose-sparing policy that selects the smallest safe regimen reaching the best achievable confidence. It attained 87.3% with zero simulator-scored toxicity at 0.63 times the median label dose. Both policies improve on the weight band in this simulation, and they trade differently: one maximises attainment, the other minimises drug used.

We make the default policy anchor on the label dose and depart from it only for a stated reason — unacceptable predicted toxicity, or predicted failure to reach the target, so that every deviation is auditable. The dose-sparing policy works by spending the margin between the label dose and the target, and therefore inherits all of the uncertainty in that target, which in this study is a quantity we specified. The point is that the decision rule carries at least as much weight as the model, and is not a technical default to be chosen silently.

Discussion

The result we regard as established is narrow and methodological: when the generative process is known and complex, a model that embeds its structure and is supervised on its latent quantities can invert it from realistically degraded observations, and both the structure and the supervision make measurable, separable contributions.

The size and ordering of those contributions is the quantitative claim, and it changed when the simulator was corrected. Supervision is worth 3.72-fold, architecture 1.54-fold. An earlier version of this work reported the reverse ordering and built an argument on it about encoding known physics as architecture rather than as training signal. That argument does not survive its own simulator’s recalibration, and we think the honest reading is narrower than either version: architecture and supervision constrain a model in different places. Supervision constrains it where data exist. Architecture constrains it everywhere, including at dose levels, sampling schedules and covariate combinations the training set never contained, which is why the solver

remains necessary for the inverse problem even though it is the smaller numerical effect. A practitioner choosing between them is not choosing between two routes to the same destination.

Relation to pharmacometric practice

The architecture is best understood as a learned, covariate-aware prior over individual pharmacokinetic parameters, standing in for the population average that conventional population modelling supplies. That reframing is the practical point. Bayesian forecasting from one or two measured concentrations is established practice [23]; its weakness in this population is that the prior is derived from cohorts that exclude the patient in front of you. Replacing the prior while leaving the inferential machinery intact is a smaller and more auditable change than replacing the whole approach, and the model supports it directly: the implementation includes maximum-a-posteriori refinement in which the network supplies the prior and measured concentrations supply the likelihood.

The fixed allometric exponents deserve emphasis for the same reason. Because they are not learned, the quantity the network estimates is exactly the residual from the conventional scaling rule. A reader who trusts allometry and distrusts the network can read the learned deviation as the claim under test, which is the form in which such a claim is most easily rejected.

On interpretability

The seven-parameter bottleneck provides something stronger than feature attribution, because the intermediate quantities are not post hoc explanations but the actual computational path. When the model predicts under-exposure it can be asked whether this is driven by clearance or by bioavailability, and the answer is the number the solver used. In the one worked case we examined in detail, a severely oedematous child on rifampicin, the model inferred a bioavailability of 0.15 and reported that no regimen in its search space reached the target — identifying absorption rather than clearance as the binding constraint, and therefore indicating that dose escalation could not resolve it. Whether such an inference would be correct in a real patient is exactly what this study cannot address; what it shows is that the architecture makes the inference legible rather than hidden.

The limits of that legibility are also visible. Permutation importance ranked sex third among all covariates, above every toxicant and laboratory value, which the simulator’s construction does not explain. An interpretable bottleneck does not make every behaviour of the surrounding encoder interpretable.

What the two failures imply

The held-out-drug failure bounds the method’s scope sharply. The descriptor branch supports interpolation within covered chemical space and nothing beyond it, and nine drugs cover very little. This is the most important open question raised here, and the obvious experiment, scaling the library by an order of magnitude and repeating the held-out-drug protocol, is well defined and has not been done.

The calibration failure generalises further than this architecture. Mechanism-constrained models guarantee physical admissibility, which is a property of individual predictions; they guarantee nothing about the honesty of the predicted distribution. We found these to be independent, and found the combination of structural correctness with overconfidence to be the most hazardous configuration, because every individual output looks reasonable. Any mechanism-constrained model whose posterior feeds a decision rule should be checked this way, and the check is cheap.

The path to a meaningful claim

Three steps separate this work from anything that could be said about a patient. First, the simulator's coefficients must be anchored to real pharmacokinetic data from undernourished cohorts rather than chosen for plausibility. Second, the model must be re-derived against that anchored simulator. Third, predicted target attainment must be validated prospectively in a real cohort.

The first step has been begun and is reported above. It was worth more than its scale suggests: comparing against four published sources exposed two structural defects that the entire unit-test suite had missed, and showed one mechanism to be threefold too strong. Both failures were invisible from inside the system, because each component behaved exactly as specified and the error lay in how they composed. That is an argument for doing this comparison early rather than treating it as a final validation step, and it is the part of this project we would advise anyone building a similar simulator to copy.

It also had consequences the first version of this paper did not anticipate. Correcting the simulator did not merely shift numbers; it reversed one argument (architecture versus supervision), eliminated one headline result (over-exposure), halved the force of another (the ablation asymmetry), and introduced a new failure case (amoxicillin) that turned out to be a target-definition problem rather than a model problem. A reader should take from this that results derived from an uncalibrated simulator are not merely imprecise — their qualitative conclusions are not safe either.

The first step is nonetheless far from complete: three drugs compared, one mechanism fitted to a single confounded study, six drugs untouched. The third step has not been attempted at all, and no amount of further work on the architecture substitutes for it.

We note the asymmetry deliberately. The machine-learning contribution here is complete and evaluated; the biomedical claim is entirely absent, and the gap between them is not a matter of degree.

What this study does not establish

This section is placed before Methods rather than after the Discussion because the constraints below govern how every preceding number should be read.

No real data were used. There were no human participants, no biological samples and no clinical measurements. Every figure in this paper describes a model's fidelity to a system of equations that we wrote.

The simulator is partly calibrated and nowhere validated. Most numerical constants were selected to be plausible rather than fitted. Three sets are now anchored to measurement: gentamicin reference clearance and volume, the toxicant potency scale, and the enteropathy coefficients governing oral absorption. That anchoring rests on three drugs and four sources, one of which is a review and one of which studied adults. Six of the nine drugs in the library have had no absolute comparison against published data, and the largest remaining error is probably in plasma-protein binding, which two separate checks independently implicate.

The coupling between nutrition and toxicant handling, the project's distinctive claim, is its least sourced component. Reviewing the literature for each mechanism found the general pharmacology well supported and the nutrition-dependent limb repeatedly unsupported. S-adenosylmethionine is the methyl donor for arsenic methylation [21], but methyl-donor *limitation under protein deficit* has no source we could verify. Iron deficiency up-regulates DMT1 and increases cadmium absorption in rats [14], but the extension to lead, manganese and to children does not. α1-acid glycoprotein is a positive acute-phase protein that binds basic drugs [12], but the resulting fall in free fraction is an inference rather than a reported finding. The organ-sparing exponent of 0.45, the mechanism external validation found inert and then repaired, has no measured source at all. The coupled axis is therefore built on mechanistic plausibility, in exactly the way the rest of the simulator is, and should not be read as better established because it is more specific.

Several pharmacodynamic targets determine their own results. Five of the nine are now sourced — the beta-lactam fT>MIC targets to Craig [25], gentamicin to Hollander et al. [26], lumefantrine to the WWARN meta-analysis [27], and fluconazole to the EUCAST technical note [28], although the fluconazole literature is not unanimous and Andes [30] gives a fourfold looser value. Four are not. The consequences are concrete rather than theoretical:

- **Gentamicin.** The toxicity threshold moves the over-exposure result from 0.0% to 12.5%, documented above. Now set to the sourced paediatric value.
- **Amoxicillin.** 50% against 40% fT>MIC moves attainment from 8% → 15% to 19% → 72%.
- **Rifampicin.** The implemented free AUC/MIC target of 38 is unsourced and is not attained by a well-nourished child at label dose. Goutelle et al. [29] report a plasma AUC0-24/MIC target of 271 for 1-log kill, roughly 54 on a free basis, which this simulator would not attain at any plausible dose.
- **Isoniazid.** The implemented target of 14 is roughly fortyfold below published values and is met with a wide margin even in severe malnutrition.
- **Albendazole.** Activity tracks the sulfoxide metabolite, which this simulator does not model, so an index on the parent compound has no clean published counterpart.
- **Paracetamol.** Carries a placeholder AUC/MIC criterion with a nominal MIC, meaningless for an analgesic. Excluded from the dosing experiment; no reported result depends on it, but it should be replaced.

The dose-sparing policy is particularly sensitive to all of this, since it operates by consuming the margin between the label dose and the target.

The dose results do not measure clinical benefit. The 69.1% to 87.6% change in target attainment was produced by scoring both arms with the same simulator that generated the training data, against pharmacodynamic targets we also specified. It demonstrates successful inversion of a known process. It is an upper bound on real-world benefit and is not evidence of it.

The model does not transfer to drugs outside its library. Held-out-drug performance was a failure, not a degradation. Nothing here supports applying the model to a compound it was not trained on.

Structural assumptions limit the simulator. Pharmacokinetics is linear and time-invariant throughout. There is no saturable protein binding, which matters for ceftriaxone; no enzyme auto-induction, which matters for rifampicin; no metabolite tracking, so pathways whose clinical significance lies in a metabolite are represented only through the parent compound; no transporters, no drug–drug interactions, and no nutritional recovery over time, although recovery is the clinical context in which such a model would most plausibly be used.

The toxicokinetics is a steady-state approximation. Real body burdens depend on intermittent exposure, growth dilution and bone remodelling, none of which are represented.

Measurement error is assumed optimistically. Anthropometric error was set at about 1.2%, which is likely below field reality for mid-upper-arm circumference, weight-for-height z-score and oedema grading.

The uncertainty correction is a single scalar. Variance recalibration repairs average coverage; it does not make the posterior correct patient by patient, and a deep ensemble would be the stronger remedy.

The model over-attributes to sex. Permutation importance ranks sex third among all covariates. Flipping sex moves the model's predicted volumes three-and-a-half to nearly five times more than it moves the simulator's, and moves five parameters the simulator does not move at all. The attribution method is reporting the model correctly; the model has learned a sex dependence the mechanism does not contain,

most likely because four other inputs are computed against sex-specific references. Auxiliary supervision on latent physiology did not prevent it.

Nothing here should inform the care of any patient. The system is a research simulator and a model trained on its output. It is not a medical device, has not been validated, and must not be used to make or influence a dosing decision.

Methods

Virtual population

Patients aged 6–59 months were drawn from a configurable mixture over five nutritional strata (adequate, mild, moderate, severe marasmic, severe oedematous). Height-for-age and weight-for-height z-scores were sampled per stratum with stratum-specific means and dispersions, and converted to absolute anthropometry through smooth analytic approximations to the WHO 2006 median growth curves. Mid-upper-arm circumference was sampled conditionally on weight-for-height z-score with independent residual variance, so that it carries information the z-score does not. Oedema grade was assigned in the oedematous stratum and inflated measured weight by approximately 3% per grade.

Fat-free mass used a paediatric body-composition relation [9] with sex-specific constants narrowed relative to the adult equation, since compositional dimorphism is small before puberty. A wasting correction raised the fat-free fraction asymptotically toward a near-fat-free bound as weight-for-height z-score fell, so that composition saturates smoothly rather than at a clipped ceiling. Total body water was taken as 76% of fat-free mass, partitioned into extracellular and intracellular compartments with an extracellular fraction rising with both oedema grade and wasting; total body water is elevated in oedematous malnutrition and falls as oedema resolves [11].

Liver and kidney mass were scaled from body surface area with an organ-sparing exponent of 0.45 applied to the weight-deficit ratio, so visceral organ mass falls sub-proportionally to body weight. The exponent itself is **not** derived from measurement and is the least supported constant in the simulator; see the list of unsourced claims. Following the correction described in External validation, surface area is computed from the size the child would have at their own height without wasting, not from actual weight. Hepatic steatosis in oedematous malnutrition increased liver mass while reducing functional hepatocyte fraction, the two being modelled separately. Cardiac output followed an allometric baseline reduced by wasting and oedema and partially compensated by anaemia; hepatic and renal blood flows were fractions of cardiac output, themselves reduced by wasting. Glomerular filtration rate was derived from kidney mass and postnatal maturation rather than from a creatinine equation [10], because serum creatinine is confounded by low muscle mass in precisely this population.

Enzyme activity factors for nine pathways were computed as multiplicative departures from a healthy reference, driven by protein deficit, a saturating function of C-reactive protein representing acute-phase down-regulation [16], micronutrient deficit, and an energy-deficit term that induces rather than suppresses CYP2E1. Unbound fraction followed the single-class protein-ratio relation [24], exact at low saturation, with albumin or α1-acid glycoprotein as the binding protein according to drug class [12]. Gastrointestinal function returned mucosal surface, paracellular permeability, gastric pH, bile-salt availability, transit slowing and intestinal CYP3A4 activity as functions of enteropathy severity [13], oedema and micronutrient status.

Toxicant exposure

Six drinking-water contaminants were modelled (arsenic, lead, cadmium, fluoride, nitrate, manganese) across six exposure settings with setting-specific log-normal source concentrations. Absorbed dose per kilogram combined water concentration, age-dependent intake per kilogram, and a nutrition-dependent absorption multiplier: divalent metals received up to a 3.5-fold increase under severe micronutrient deficiency representing transporter up-regulation [14], and all solutes received a paracellular increment under enteropathy. Body burden used one-compartment chronic accumulation toward steady state over an exposure duration proportional to age, with elimination scaled by a nutrition-dependent detoxification capacity representing methyl-donor and cysteine limitation [21]. Burdens acted on disposition through saturating Hill relations producing tubular toxicity, hepatocellular injury, mucosal injury, protein-binding displacement, haem impairment [15] and oxidative stress, together with per-enzyme modifiers.

Pharmacokinetic derivation and solution

Hepatic clearance used the well-stirred model [7] with unbound fraction in blood and an intrinsic clearance proportional to liver mass and weighted enzyme activity. Renal clearance was filtration of unbound drug. Distribution volume used a reduced tissue-composition model in which plasma, extravascular extracellular fluid, intracellular fluid and adipose tissue contribute according to lipophilicity and binding. Oral bioavailability combined fraction absorbed (mucosal surface, bile-salt dependence, gastric pH sensitivity, paracellular contribution for small hydrophilic molecules) with gut-wall and hepatic first-pass extraction. Two constants per drug, fitted once against a declared reference clearance and volume in a well-nourished reference child, set absolute magnitude; all covariate dependence is mechanistic.

The resulting system is an Erlang transit-absorption chain [8] feeding a two-compartment disposition model. It is solved exactly by matrix exponential, with zero-order infusions handled through an augmented-matrix construction so that bolus, infusion and oral administration share one code path. State vectors are padded to a fixed width so that drugs with different transit-chain lengths share a batched implementation.

Dataset

12,000 virtual patients were simulated, each receiving three drugs drawn without replacement from the nine-drug library, giving 36,000 concentration–time records. Dose multipliers spanning 0.5–2.5× and interval multipliers spanning 0.5–2× the label regimen were sampled per record, so the dose–response surface is observed rather than a single point on it. Each record comprises 50 observations across five dosing intervals.

Residual error was log-normal with a coefficient of variation of 0.18 plus a small additive term near the limit of quantification; values below that limit were reported as zero and masked from the likelihood rather than clamped, which would otherwise bias the terminal slope. Sampling schemes were rich (15%), intermediate (30%) or sparse (55%); sparse records retain two to four opportunistic samples biased toward the terminal interval. Laboratory values were independently masked at field-plausible rates, with α1-acid glycoprotein masked most often.

Splits are by patient, never by record. Three split regimes were constructed: random patient (70/15/15, giving 25,200 / 5,400 / 5,400 records), held-out exposure setting (trained without industrial-runoff water; 27,540 train / 3,600 test), and held-out drug (trained without rifampicin and gentamicin; 23,824 train / 7,974 test).

Architecture and training

Per-block encoders map anthropometry, laboratory values with their missingness mask, water chemistry and drug descriptors to a 128-dimensional representation; the drug embedding produces FiLM scale and shift

parameters applied multiplicatively to each patient block, plus a scalar gate per block [20]. Missing laboratory values are replaced by a learned per-feature token rather than a zero. The fused representation feeds a parameter head whose output is a drug-level baseline from descriptors, a fixed allometric term, and a bounded patient deviation (tanh-limited to about a tenfold swing). Volumes and clearances are exponentiated; bioavailability and unbound fraction pass through a sigmoid. A heteroscedastic head emits residual scale; an auxiliary head predicts 28 latent physiological quantities.

The loss is a log-normal negative log-likelihood on observed, non-censored concentrations, plus weighted mean-squared error on log parameters and standardised auxiliary targets. The black-box baseline shares all encoders and the fusion trunk, additionally receives the regimen block (which the hybrid deliberately excludes), and regresses the 50-point curve directly.

Training used AdamW, batch size 384, peak learning rate 3×10^{-3} with 5% linear warm-up and cosine decay, gradient-norm clipping at 2.0, 30 epochs, early stopping with patience 12. The structural head was initialised at the empirical median of the centred log parameters so that the first forward pass begins at the right order of magnitude. Model selection used log mean-squared error against the noise-free curve on validation, not the noisy likelihood. Normalisation statistics were computed on training data only, with laboratory statistics computed over observed values only.

Evaluation

Concentration accuracy is reported as log-RMSE, median and 90th-percentile absolute fold error, and the fraction within twofold, computed against the noise-free simulated curve at all time points. Parameter recovery is reported as R^2 on the log scale, with absorption rate excluded for intravenous records. Permutation importance permuted each feature within its observed marginal with two repeats, permuting laboratory values jointly with their mask. Calibration was assessed separately for residual and epistemic components against their respective targets, with the epistemic scale fitted on validation and applied to test. Monte Carlo dropout used 16–40 samples with normalisation layers held in evaluation mode.

Dose selection evaluated a 19-point logarithmic dose grid under Monte Carlo dropout with the fitted uncertainty scale, computing probability of target attainment and of exceeding the toxicity threshold at each candidate; the dosing interval was searched over three multipliers when no dose satisfied both constraints. The anchored policy retains the label dose unless predicted toxicity is unacceptable or predicted attainment falls below 90% confidence.

External calibration and validation

Four published sources were retrieved and read; values were transcribed into a reference module carrying each entry’s citation, population description, confounders and a verification flag distinguishing values read from the primary study from those reported by a review. Quantities that could not be obtained were recorded as absent rather than estimated, and one source could not be retrieved at all and is therefore missing rather than approximated.

For each comparison, a virtual cohort was constructed to match the study’s described nutritional phenotype mix and age range, and its weight distribution was checked against the published one before the comparison was accepted. Dosing followed each study’s actual regimen, including the fixed-tablet artemether-lumefantrine schedule and sampling at 168 hours for the day-7 measurement. Parameters were compared as fold differences and against published variability intervals where reported; cohort contrasts were compared as ratios, which are independent of the reference-anchoring constants and therefore test the covariate model rather than the calibration.

Three sets of constants were adjusted in response. Gentamicin reference clearance and volume were set to published normal paediatric values. Toxicant half-maximal-effect constants were re-expressed as multiples of each toxicant's guideline-equivalent steady-state burden, with multiples between 4 and 10. The enteropathy coefficients acting on mucosal surface and bile-salt availability were fitted by grid search against the published day-7 lumefantrine ratio, averaged over three independent cohort draws; single-draw fitting was tried first and abandoned after it produced an apparently excellent match that did not reproduce across seeds. No other constant was changed, and the functional forms were left intact throughout.

Verification

Because the simulator is the ground truth, its correctness is load-bearing. Forty-seven automated tests cover the implementation, weighted toward physical rather than functional correctness: conservation of mass over a long horizon; the identity AUC = F·D/CL to within 0.1%; exact dose proportionality; superposition of repeated doses against explicitly summed single-dose responses; agreement between the padded batched solver and a reference implementation to 10^{-9} relative error; agreement between the differentiable PyTorch solver and the NumPy ground truth to below 10^{-9}; agreement between differentiable and reference exposure metrics; and the direction of every mechanistic covariate effect the simulator claims to encode. A unit-mismatch error in the differentiable time-above-threshold metric was detected by the metric-agreement test and corrected before the dose experiments were run.

Software

Python 3.13, PyTorch 2.14, NumPy, SciPy. All experiments ran on two CPU cores; the full pipeline, including dataset generation, nine training runs, evaluation and figures, completes in approximately 2.5 hours.

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Claims still lacking a verified source

Nine assertions in the Introduction or Methods have no source in the list above. Each should be sourced, softened to a stated modelling assumption, or removed.

Claim	Status
Visceral organ mass falls **sub-proportionally** to body weight (the 0.45 sparing exponent)	No measured source. Published ultrasound data confirm organs are smaller in malnourished children but report correlations, not allometric slopes. The route not yet followed is the organ-mass references used by [5].
Extracellular water expands with **reciprocal intracellular depletion**	[11] supports elevated total body water only, not the compartment split.
Creatinine-based GFR is **biased upward at low muscle mass**	[10] gives the paediatric equation but does not establish the bias.
EED specifics: villous blunting, raised gastric pH, bile-salt deficiency	[13] supports EED as inflammation, reduced absorption and reduced barrier function, not these three mechanisms.
AGP rise **lowers unbound fraction of basic drugs**	[12] supports the acute-phase rise and basic-drug binding; the combined claim is an inference.
DMT1 up-regulation extends to **lead and manganese**, and to **children**	[14] covers cadmium in rats.
Arsenic methylation is **methyl-donor limited under protein deficit**	[21] supports SAM dependence only.
Cadmium nephrotoxicity **reduces glomerular filtration**	[22] supports tubular dysfunction at borderline significance; it did not measure GFR.
PK/PD targets for rifampicin, isoniazid, albendazole; the paracetamol placeholder	See the dosing section. Three of these determine their own results.

The pattern matters more than the count. For five separate mechanisms the general pharmacology is well sourced and the nutrition-dependent limb is not: SAM is the methyl donor for arsenic methylation but

methyl-donor limitation under protein deficit is unsourced; iron deficiency raises DMT1 and cadmium uptake in rats but the extension to lead, manganese and children is not; α1-acid glycoprotein rises in inflammation and binds basic drugs but the resulting fall in free fraction is an inference; cadmium raises tubular injury markers but a reduction in filtration is unsourced; organs are smaller in malnutrition but sub-proportional scaling is unsourced. The coupling between nutritional status and toxicant handling, this project's distinctive claim, therefore rests on mechanistic plausibility rather than on measurement, in exactly the way the manuscript says the rest of the simulator does.